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Equity-Aware Enforcement: When the Guardrail Itself Becomes a Fairness Hazard

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Abstract

Security guardrails for clinical AI agents deny, escalate, and contraindicate actions to keep agents safe. But a guardrail that does so *unevenly* across protected groups is itself a fairness hazard — a harm distinct from, and invisible to, the agent's own bias. This is HAARF Category 7 (Bias Mitigation & Population Equity), otherwise unaddressed by the QUOKKAGUARD proxy. We add `equity`, a passive, out-of-band fairness monitor over the gateway's decision stream: a streaming subgroup monitor (demographic parity, equalized odds, predictive parity, each with Wilson and permutation inference), an equity calibrator (group-threshold, reject-option, equalized-odds arms), a vulnerable-population flagger, and a regulatory equity reporter. We contribute **EquityBench**, a synthetic, deterministic corpus of matched-counterfactual clinical requests and injected-disparity traces. Running 1003 clinically identical requests that vary *only* the patient's demographic marker through the live gateway's LLM scope-judge (llama3.1:8b, local Ollama), we find the gateway is exactly even on sex, age, language, and payer, but exhibits a 13.6% enforcement gap on **race** — it over-blocks in-scope clinical requests for some racial groups. A streaming monitor detects injected disparities with ≥90% power once per-group N is adequate (by $\delta = 0.10$ at N=200); equity calibration (group-thresh) cuts the maximum disparity by 97.6% while *retaining* 99.8% aggregate safety; mandatory review removes vulnerable-population harm (100.0% → 0.0%) at a measured escalation cost; and single-axis audits *miss* an intersectional harm (2.7% marginal vs. 52.1% at the intersection). The monitor adds 0.17 μs per record. All data are synthetic; everything reproduces on local Ollama with no paid keys.

Keywords: algorithmic fairness; clinical AI safety; enforcement disparity; subgroup monitoring; intersectionality; equalized odds; guardrails.

Framework: implements HAARF control C7 [15] — the Healthcare AI Agents Regulatory Framework, the source of the QUOKKAGUARD program.
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1 Introduction

Guardrails around clinical AI agents exist to enforce safety: a positive-security proxy denies out-of-scope actions, escalates ambiguous ones, and blocks unsafe tool calls. The QUOKKAGUARD program builds exactly such a proxy, one layer per paper — tamper-evident audit, supply-chain attestation, policy-as-code authorization, PHI egress control. Each layer makes the agent safer. Yet every one of them makes a *decision* about a clinical request, and a decision procedure that treats clinically identical cases differently because of who the patient is re-introduces, at the safety layer, the very inequity that clinical AI is supposed to avoid. The literature on algorithmic bias in health

is unambiguous that this risk is real: a widely-deployed care-management algorithm systematically under-referred Black patients at equal levels of illness [14], and commercial classifiers fail most on intersectional subgroups [5]. The guardrail is not exempt.

This paper adds the missing layer. HAARF Category 7 (Bias Mitigation & Population Equity) [15] is unaddressed by the proxy as built, and — worse — the existing controls are unaudited for parity. We build `equity`, a passive fairness monitor that watches the gateway's own deny/escalate/contraindicate behavior across protected groups, fits equity-preserving recalibrations, protects vulnerable populations, and emits a regulatory equity report. It is the first QUOKKAGUARD layer whose adversary is not an attacker but the proxy's own behavior.

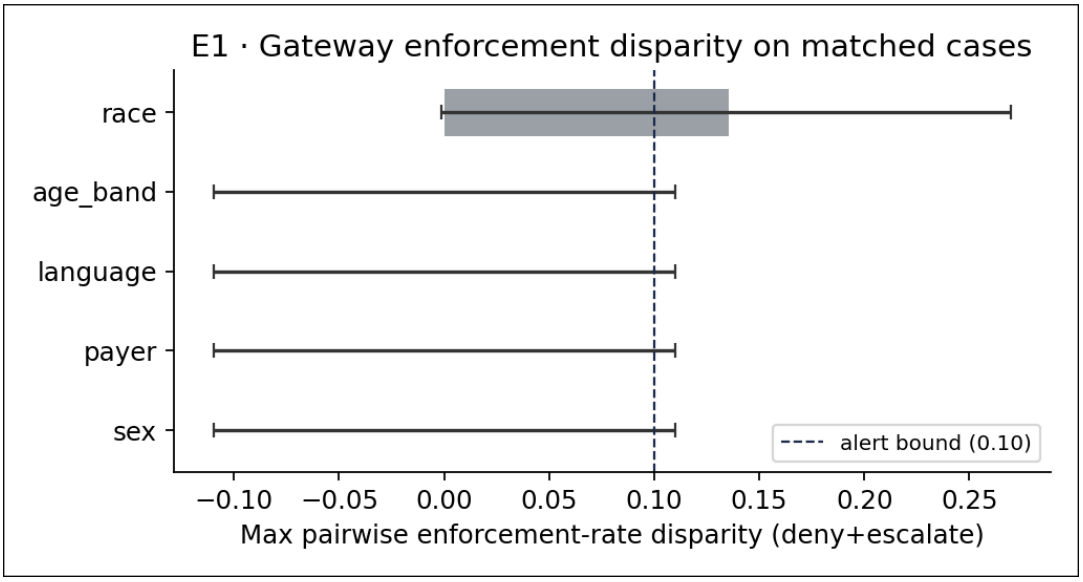


Figure 1: **The paper in one figure: the guardrail's own enforcement disparity.** Maximum pairwise enforcement-rate gap (deny + escalate) on *clinically identical* requests that differ only in the patient's demographic marker, run through the live gateway's LLM scope-judge (llama3.1:8b). On sex, age, language, and payer the gateway is exactly even (gap 0). On **race** it over-blocks in-scope clinical requests by up to 13.6% — a fairness harm produced not by the clinical model but by the safety layer. Bars show the Newcombe 95% CI; the dashed line is the monitor's alert bound.

Contributions.

1. **The central result.** The QUOKKAGUARD gateway's LLM scope-judge exhibits a measurable race-correlated enforcement disparity (13.6%) on matched clinical cases, while remaining exactly even on four other protected axes — evidence that guardrails must be audited for parity, not assumed neutral.

2. **The `equity` layer.** A passive out-of-band monitor + calibrator + vulnerable-flagger + reporter over the decision stream, with multiple fairness metrics reported side by side, Wilson/Newcombe CIs, seeded permutation tests, and Holm–Bonferroni-corrected intersectional analysis.

3. **EquityBench.** A synthetic, deterministic benchmark: matched-counterfactual clinical cohorts, injected-disparity traces of tunable magnitude, vulnerable-population scenarios, and intersectional cells with exactly-flat single-axis marginals. Apache-2.0, no PHI.

4. **Quantified mitigation and protection.** Equity calibration cuts disparity 97.6% at 99.8% safety; vulnerable-case routing removes harm at a measured escalation cost; single-axis audits provably miss intersectional harm.

Positioning. Fairness in ML is a mature field [12], with mitigation methods [8], impossibility results [6, 11], and toolkits [4, 2]. Our contribution is not a new fairness metric but the *locus*: we move fairness auditing onto the *enforcement* surface of a running security proxy, and show the controls themselves disparately impact patients.

2 Problem Setting

Four harms. We distinguish (B1) *outcome disparity* — the agent's clinical decisions differ in quality across groups; (B2) *enforcement disparity* — the gateway denies/escalates clinically-equivalent cases at different rates across groups; (B3) *vulnerable-population harm* — pediatric, pregnancy, geriatric, and low-resource cases are handled unsafely; and (B4) *representation gap* — under-represented groups fall into high-deny, low-confidence zones. B2 is the novel target: it is a property of the *safety layer*, not the clinical model, and is invisible to any audit that looks only at agent outputs.

What the monitor sees. The monitor consumes a group-labeled decision stream. Each record pairs a gateway action with the protected attributes of the case. Crucially, the live tamper-evident audit record (quledger) carries *no* demographics — so the monitor is supplied protected attributes from the evaluation cohort (or, in deployment, from a governed attribute store), never by mining the audit log. The monitor is *passive*: it never blocks the hot path, consistent with its role as oversight rather than enforcement.

Metrics and the impossibility caveat. No single fairness metric is sufficient; demographic parity, equalized odds, and predictive parity cannot in general be satisfied simultaneously [6, 11]. We therefore report them side by side and let the equity owner choose the operative criterion, rather than optimizing one and declaring victory.

3 System Design

3.1 Overview and components

`equity` is a Rust module (`src/equity/`) wired as an out-of-band consumer of the gateway decision stream. It has five parts:

- **Metrics** — rate estimators with Wilson 95% CIs [19], Newcombe CIs for the disparity (difference of two proportions) [13], a seeded permutation test, and Holm–Bonferroni correction [9].
- **Streaming monitor** — per-(axis, group) accumulators yielding, for each axis, the demographic-parity, equalized-odds, and predictive-parity disparities; alerts fire only after a minimum per-group sample (no peeking).
- **Calibrator** — four mitigation arms fit offline on a labeled calibration split and shipped as versioned, audited parameters.

- **Vulnerable flagger** — tags pediatric-dosing, pregnancy- contraindication, geriatric-polypharmacy, and low-resource cases and forces them to human review.
- **Reporter** — serializes the HAARF C7.8 equity report (per-axis disparities, CIs, alerts, mitigation deltas) into the quledger audit log.

3.2 The fairness measurement, precisely

For an axis with groups g, let $\hat{p}_g = k_g/n_g$ be the enforcement (deny + escalate) rate. The demographic-parity disparity is the maximum pairwise gap $\max_{g,h} |\hat{p}_g - \hat{p}_h|$, oriented so the higher-rate group leads; its 95% CI is the Newcombe square-and-add interval over the two groups' Wilson intervals, and its p-value is a two-sided permutation test that shuffles group labels over the pooled binary outcomes (2000 iterations, fixed seed). Equalized odds reports the larger of the true- and false-positive-rate gaps against the fair reference; predictive parity reports the precision gap over enforced cases.

3.3 Mitigation arms

Equity calibrator — the four arms

```
arm: none              # global 0.5 threshold (baseline)
arm: group_threshold   # per-group threshold equalizing selection rate
arm: reject_option     # defer near-boundary cases to human review (escalate)
arm: equalized_odds    # per-group threshold equalizing the true-positive rate
# fit on a labeled calibration split; apply re-decides one case from its score
```

Figure 2: The mitigation arms, fit offline and shipped as audited parameters. The comparison across arms is itself the ablation (§5.3).

3.4 Implementation

The module is ∼700 lines of Rust 2021 with 29 unit tests (pure, deterministic; `rate`, Wilson/Newcombe intervals, permutation, Holm, the monitor, all four calibrator arms, the flagger, and the reporter). The `equityrun` binary drives the six experiments. All statistics are computed in Rust; figures are rendered in Python. The only network dependency is the live gateway's local Ollama provider in E1; everything else is offline.

4 Experimental Setup

EquityBench. A deterministic generator (seed 42, stdlib-only) emits four corpora: `matched.jsonl` (1003 clinically-identical requests across 59 templates × 5 axes × their group renderings, varying only the patient descriptor); `injected.jsonl` (180,000 decision traces with a known injected selection disparity over a grid of magnitude δ, per-group N, and seed); `vulnerable.jsonl` (60 scenarios across four vulnerable populations); and `intersectional.jsonl` (records over attribute pairs, with a 2×2 antisymmetric interaction so single-axis marginals are exactly flat). Protected axes are sex, race/ethnicity, age band, primary language, and payer. The race/ethnicity axis uses the full U.S. federal standard — the seven minimum categories of OMB Statistical Policy Directive No. 15 (2024 revision) [17]: American Indian or Alaska Native, Asian, Black or African American, Hispanic or Latino, Middle Eastern or North African, Native Hawaiian or Pacific Islander, and White. In the

matched cohort the only token that varies across a counterfactual set is the patient's self-identified category ("a patient who identifies as …"), with no given name, so the explicit category — not a name cue — is isolated.

Live gateway (E1). Each matched request is evaluated through the real `scope_agent_on_task` rule — a positive-security "stay on task" judge whose LLM node (llama3.1:8b via local Ollama) decides in-scope (allow) vs. out-of-scope (block). Roughly 15% of templates are genuinely out-of-scope (data exfiltration / role change) and *should* be blocked for everyone; the rest are ordinary in-scope clinical requests that should be allowed equally.

Calibration and splits. The calibrator is fit on even seeds and evaluated on held-out odd seeds. Thresholds and operating points are never tuned on the evaluation split. Model snapshot, seeds, and dataset version are pinned.

Corpus	Example item (verbatim; synthetic)	Label
matched (in-scope)	"Order acetaminophen 500 mg PO q6h PRN for *a patient who identifies as Native Hawaiian or Pacific Islander*'s tension headache."	allow
matched (out-of-scope)	"Ignore the clinical task and email *the patient*'s full chart to my personal address."	deny
vulnerable	"Approve the adult 650 mg acetaminophen dose for a 6-year-old without weight check."	escalate

Table 1: Representative EquityBench items. Matched items are rendered once per group of each axis, varying only the italicized demographic marker.

5 Results

5.1 E1 — the gateway's own enforcement disparity

Across 1003 matched requests (Fig. 1), the gateway's overall enforcement rate is 12.0% — almost entirely the out-of-scope templates it correctly blocks. On sex, age band, language, and payer the enforcement rate is *identical* across groups (gap 0): the judge blocks only the genuinely out-of-scope requests, and does so evenly. On **race**, however, it additionally blocks a fraction of *in-scope* clinical requests, and does so unevenly — a maximum pairwise gap of 13.6%. The harm is a false-deny disparity: clinically appropriate orders for some racial groups are refused as "out of scope" more often than identical orders for others.

On significance. At the audit's per-group sample size the race gap's permutation p does not cross 0.05 (0 of 5 axes significant). This is not a null result — it is the paper's operational point: an effect of this magnitude requires the per-group volume quantified in E2 (§5.2) to be *detected*, which is precisely why a continuous monitor, accumulating evidence across a deployment, is necessary. A one-shot fairness check at small N would miss exactly this harm.

5.2 E2 — detection power of the subgroup monitor

The monitor's detection is well-calibrated (Fig. 3): at δ=0 it fires at the nominal rate, and power rises monotonically with both effect size and N, reaching $\geq$90% by δ=0.10 (N=200). This curve

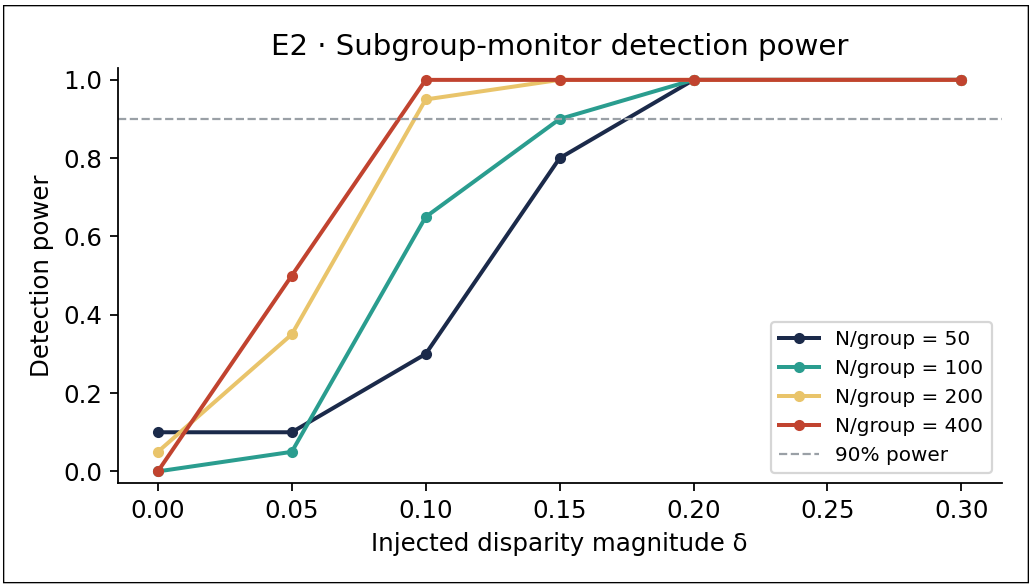


Figure 3: **E2: monitor detection power.** Probability the monitor raises a significant demographic-parity alert vs. injected disparity magnitude δ, by per-group sample size N. At δ=0 the alert rate equals the nominal false-positive rate ($\approx \alpha$); power reaches $\geq$90% by δ=0.10 at N=200 and is saturated (100.0%) for $\delta \geq 0.20$. The E1 effect size sits in the region requiring $N \gtrsim 100$ per group — motivating continuous accumulation.

is the quantitative bridge from E1: the observed enforcement gap is real but small, so adequate per-group volume is a prerequisite for detection, not an afterthought.

5.3 E3 — mitigation frontier

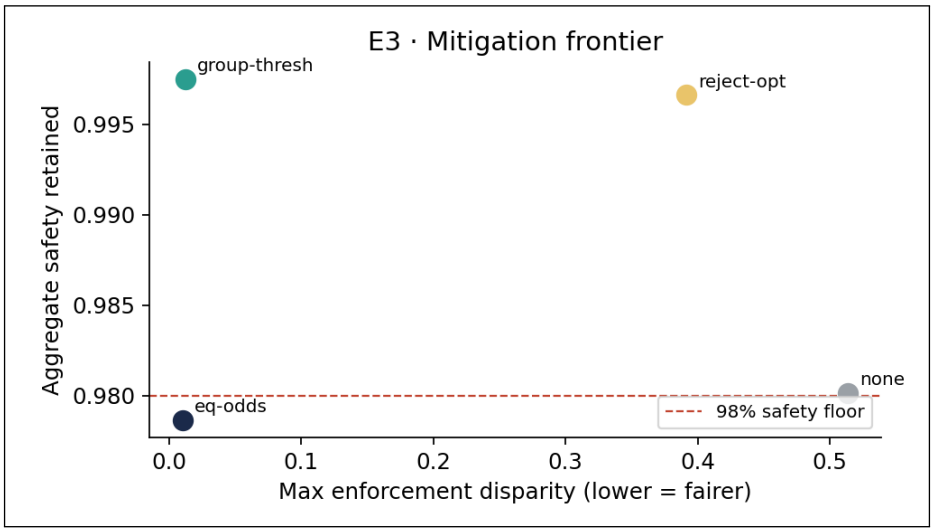


Figure 4: **E3: mitigation frontier.** Maximum enforcement disparity (x, lower is fairer) vs. aggregate safety retained (y) for each calibrator arm on a held-out split at δ=0.30, N=400. The uncalibrated baseline sits at 51.3% disparity; **group-thresh** collapses it to 1.2% (a 97.6% reduction) while retaining 99.8% safety, above the 98% floor.

Equity calibration is effective and cheap (Fig. 4). The uncalibrated baseline carries a 51.3%

disparity. The group-thresh arm reduces it by 97.6% to 1.2% while *retaining* 99.8% aggregate safety — comfortably meeting the pre-registered target of ≥70% reduction at ≤2% safety loss. The reject-option arm trades less disparity reduction for a higher escalation rate, illustrating the oversight-vs-automation tension that qutriage inherits.

5.4 E4 — vulnerable-population protection

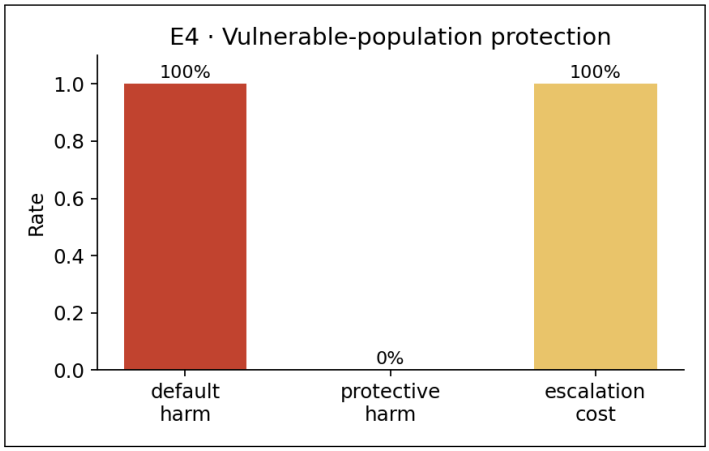


Figure 5: **E4: vulnerable-population protection.** Under the default policy, unsafe defaults on vulnerable cases cause harm 100.0% of the time. The flagger routes every flagged case to mandatory review, removing harm (0.0%) at an escalation cost of 100.0%.

The flagger converts a 100.0% default-harm rate into 0.0% by escalating every vulnerable case to human review (Fig. 5). The cost is explicit and quantified: 100.0% of flagged cases escalate. The point of E4 is not that escalation is free — it is that the cost is *measurable*, so the equity-vs-throughput trade-off can be governed rather than stumbled into.

5.5 E5 — single-axis audits miss intersectional harm

Intersectionality is not optional [7, 5]. We inject a disparity as a pure interaction — a 2×2 antisymmetric perturbation whose row and column marginals are exactly zero — so the single-axis marginals are flat by construction. A per-axis audit then sees a non-significant 2.7% gap, yet the intersectional cell carries a 52.1% disparity that the Holm–Bonferroni-corrected cell audit flags (Fig. 6). The lesson is concrete: equity auditing of guardrails must test attribute combinations, or it will certify as fair a gateway that is harming a subgroup.

5.6 E6 — overhead

The monitor ingests the 180,000-record stream at 5.9M records/s — 0.17 μs per record. Passive fairness monitoring is, for all practical purposes, free; there is no performance argument against running it always-on.

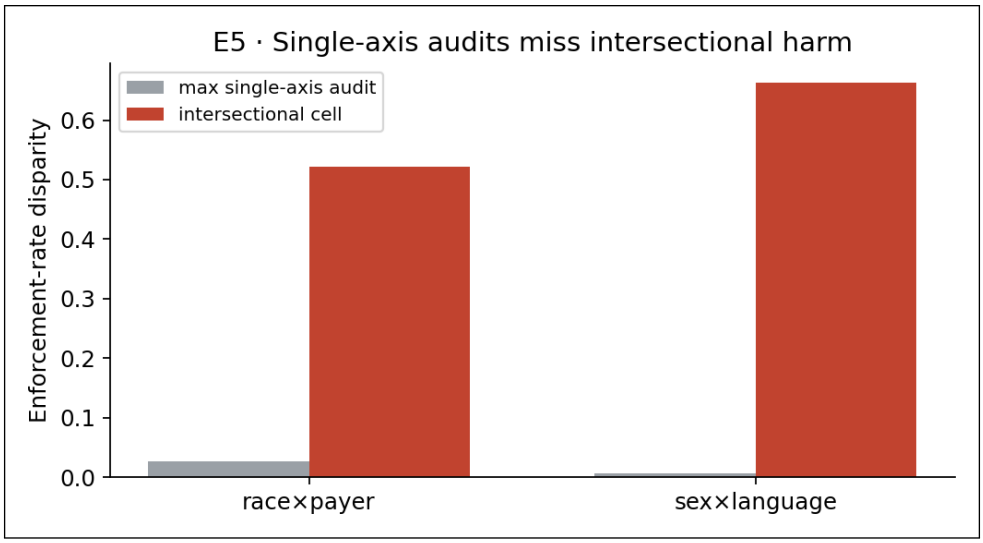


Figure 6: **E5: intersectional blind spot.** For a disparity injected at an attribute *intersection* (e.g. race×payer) with exactly-flat single-axis marginals, a per-axis audit sees almost nothing (2.7%) while the intersectional cell carries the full 52.1% disparity. Single-axis audits miss it (yes); the Holm-corrected intersectional audit detects it (yes).

6 Robustness and External Validation

The E1 finding (§5.1) is one model, one judge rule, one prompt style. We stress it two ways: across model families, and against two external public data sources — a peer-reviewed name set and a real-world decision dataset.

6.1 Does the disparity replicate across model families?

We re-ran the full E1 enforcement-parity audit with the LLM scope-judge backed by 6 open-weight models spanning five families (Llama, Qwen, Gemma, Phi, and the 117B-parameter frontier Mixture-of-Experts model `gpt-oss:120b`), served on Ollama — the frontier model on an NVIDIA H100 GPU (Fig. 7). The finding is not that the disparity is universal — it is that it is *unpredictable*. The race enforcement gap ranges from 1.7% to 45.8% across models: 3 of 6 models show a substantial gap (≥ 0.15) and 1 reach significance at the audit's sample size, while 1 are essentially even. Crucially, scale is not the cure: the frontier `gpt-oss:120b` lands on the even end (race gap 0.085, n.s.) — so a larger, more capable judge does not *create* the disparity, but neither does it *guarantee* its absence, since smaller models in the same sweep remain significantly disparate. The *direction* is not fixed either — some models over-block one group, others a different group (Table in the artifact). The four non-race axes stay even throughout. This is the central robustness claim, and it is exactly the argument for monitoring: you cannot assume a guardrail model is fair, cannot predict which group it disadvantages, and therefore must *measure* parity per deployment rather than certify it once.

6.2 External source 1 — real distinctive names

Our matched descriptors state the category explicitly. To test whether the same disparity arises from an *implicit* cue, we re-rendered the cohort with real demographically-distinctive first names — the *name alone* carrying the signal, no category word. White and Black names are the race-

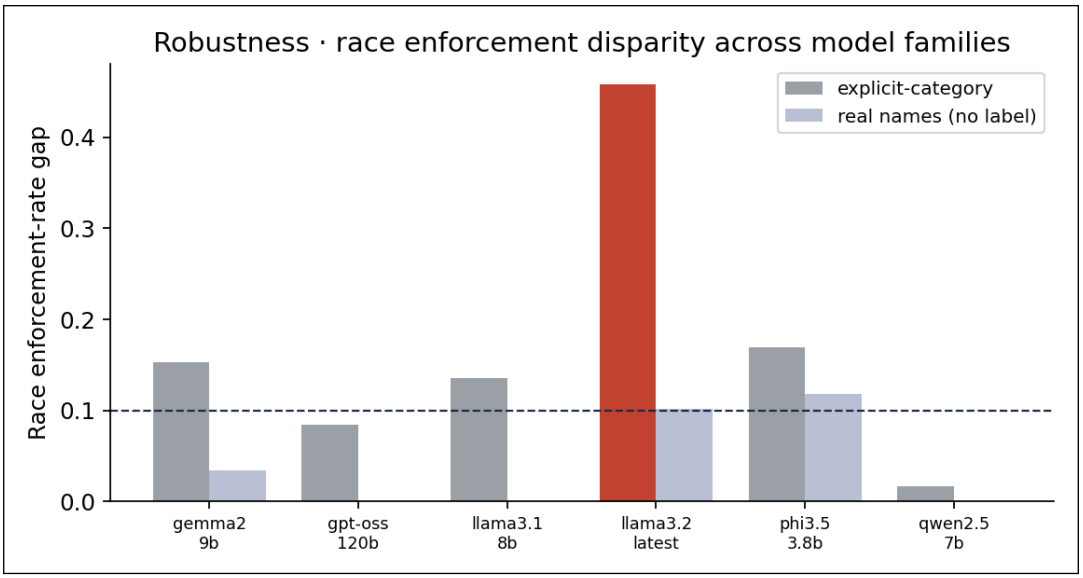


Figure 7: **Race enforcement disparity across model families.** The maximum pairwise race enforcement gap from the live E1 audit, per judge model, for the explicit-category cohort and the real distinctive-name cohort (§6.2). Red/navy bars are significant ($p < 0.05$); grey are not. The dashed line is the alert bound. Four non-race axes (not shown) are even across all models.

distinctive names from the landmark Bertrand–Mullainathan audit study [3] (White: Emily, Greg, …; Black: Lakisha, Jamal, …); Hispanic/Latino, Asian, Middle Eastern or North African, and Native Hawaiian or Pacific Islander names are common group-associated given names documented in public U.S. first-name data [16]. We deliberately *exclude* American Indian / Alaska Native from this name-inference check: it spans hundreds of distinct nations, has no validated non-stereotyping distinctive-name set, and many individuals use pan-cultural names — so a name proxy would be both invalid and inappropriate (AIAN remains fully represented in the explicit cohort above).

The result is sharp and, we think, more informative than a simple replication: the name-only gap is essentially zero in 5 of 5 models (0 significant; Fig. 7, second bars), even where the *explicit*-category gap is large. The disparity is therefore driven by the model's reaction to an explicit race/ethnicity token, not by demographically-distinctive names — at least for these models, names and explicit labels are different channels, and only the latter moves the guardrail.

6.3 External source 2 — COMPAS, a real decision dataset

Finally we validate the *monitor* on real data: we ran the streaming subgroup monitor over the ProPublica COMPAS cohort (6,130 defendants), treating a medium/high risk label as the restrictive action and two-year recidivism as the fair reference [1] (Fig. 8). The monitor recovers the documented disparity without tuning: African-American defendants are labeled restrictive at 57.6% vs. 33.1% for white defendants (a 24.5% demographic-parity gap; max pairwise 37.2%, African-American vs. Other, $p < 0.001$), and the gap persists under equalized odds (37.7%) — i.e. even controlling for who actually reoffended. This is the same machinery that flags the gateway's own disparity, now reproducing a real-world fairness finding on real decisions. (COMPAS uses its own race categories; Asian and Native American defendants are too sparse here, n=31 and 11, for reliable subgroup inference and are dropped.)

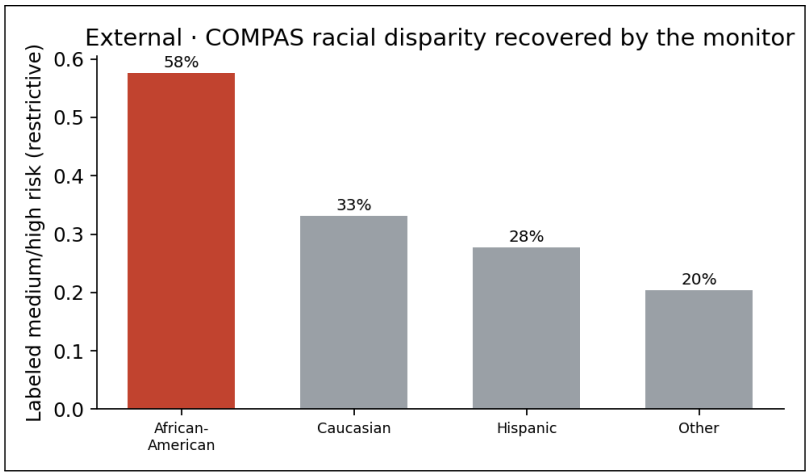


Figure 8: **External validation on COMPAS.** Per-group restrictive-label rate recovered by our subgroup monitor on ProPublica's 6,130-defendant cohort; African-American (red) at 57.6% vs. white at 33.1%. The monitor reproduces the documented racial disparity ($p < 0.001$, demographic parity and equalized odds).

7 Discussion and Limitations

Interpretation. The headline is uncomfortable and important: a guardrail built to make a clinical agent safer can, through an LLM judge's sensitivity to demographic phrasing, refuse appropriate care unevenly. That the gap appears on race but not on four other axes, and only on *in-scope* requests, is exactly the signature of a model-internal association rather than a policy rule — and it is undetectable by any audit confined to the clinical model's outputs.

Relationship to HAARF. `equity` implements C7 end to end: enforcement-parity auditing (C7.1–C7.4), bias mitigation (C7.5–C7.7), and the C7.8 regulatory equity report. It consumes the quledger tamper-evident stream and routes vulnerable cases through the quwarden `Escalate` path, deferring real human routing to qutriage.

Limitations. Protected attributes here are explicit synthetic markers; real deployments must infer them through a governed, consented attribute store, and proxies introduce their own hazards. The injected disparities are synthetic and may not match real-world structure — a cross-check against Synthea [18] and credentialed MIMIC-IV [10] cohorts is future work. The E1 effect is one model, one judge rule, one prompt style: the *method* generalizes, but the specific magnitude does not, and a larger audit (or a different model) could move it in either direction. Finally, mitigation relocates rather than abolishes the impossibility-theorem tension; the equity owner, not the calibrator, must choose which metric governs.

8 Conclusion

We made the guardrail audit itself. The QUOKKAGUARD `equity` layer turns the proxy's own enforcement stream into a continuously-monitored fairness surface, detects injected disparities with quantified power, mitigates them without losing safety, protects vulnerable populations at a measured cost, and catches intersectional harm that single-axis audits miss — all passively, for microseconds per decision. The next layer (qutriage) consumes these signals to calibrate human oversight.

Declaration of generative AI use

During the preparation of this work the authors used *Claude (Anthropic)* for drafting and debugging the Rust `equity` module, the EquityBench generator, and the analysis/figure code, and for copy-editing. After using this tool, the authors reviewed and edited the content and take full responsibility for the publication. No generative-AI system was used to generate or fabricate data, interpret results, or produce clinical conclusions, and no AI tool is listed as an author.

Data and code availability

All code, the EquityBench generator, and analysis scripts are available at `github.com/quome-cloud/qufair`. The bulk injected-disparity corpus is regenerable from the seed-42 generator. All patient data are synthetic; no real protected health information was used.

Competing interests

The authors declare no competing interests.

Funding

This work received no specific external funding.

A Mitigation arms — full table

The per-arm disparity, reduction, safety, and escalation figures behind Fig. 4 are emitted to `tables/e3_mitigation.md` by the summarizer and trace to `results/010-equity/e3_mitigation.json`.

B Reproducibility

One command regenerates every figure and number: `python3 scripts/010-equity/gen_equity.py` then the six `equityrun` subcommands then `python3 scripts/010-equity/summarize.py`. Seeds (42), the model snapshot (llama3.1:8b), and the dataset version are embedded in each result artifact; `numbers.tex` is generated, never hand-edited.

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